

Is the Labour Theory of Value Right After All?

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Why It Matters to Confirm the Validity of the Labour Theory of Value

The economic form we live in is the capitalist commodity economy. In this society, everyone produces for money. But since the gold standard left the stage of history, the production cost of fiat currency is negligible: money itself is only a symbol representing a certain "value." What that value actually is — opinions differ. I will not go into the mainstream economic story, because the subjective theory of utility cannot be empirically verified; it can only be a theoretical conjecture (see this online translation^[1]). Yet the labour theory of value advocated by Marxist economics, for its part, has always been in a state where it is true if you believe it and not if you do not. What is more, the very account in Volume III of Capital of the transformation of values into prices of production has in fact dissolved the labour theory of value itself. The neo-Ricardians have shown that final prices can bypass labour time entirely, being determined directly by the wage rate and the technical composition of commodity production, as in the following paper^[2]:

The crux is that without the labour theory of value, the historical fate of capitalism cannot be discussed on the economic plane. A socialist movement that seeks to replace capitalism would then have to rest its case on ethical and political criticism alone, lacking the most fundamental economic criticism. If one cannot predict, on economic grounds, that capitalism will fall into stagnation, then — in Marx's words, "no social order ever perishes before all the productive forces for which there is room in it have developed; and new, higher relations of production never appear before the material conditions of their existence have matured in the womb of the old society" — as long as capitalism does not stagnate and can still drive economic growth, capitalism can and should continue to exist.

This is because the fundamental motive force of capitalist development is profit. When the rate of profit is high, capital expands reproduction and drives the economy forward; when it is low, capital reduces investment and drags the economy down. Marxist economics has put forward the historical tendency of the rate of profit to fall. On non-Marxist views of value, one cannot see this historical limit of profit tending toward stagnation, and so must seek help outside the inner structure of the economy — climate change, for instance. This is why Marxists pay ever more attention to climate change. Of course, we do not say the climate question is unimportant — it is very important — but the climate question can constitute only an ethical critique of capitalism, not an economic one. In fact, as long as profit is assured, even for a decarbonised green economy there will always be capitalists willing to cooperate.

Hence, confirming the validity of the labour theory of value is a foundational, decisive theoretical question for scientific socialism. Of course, by "proving the labour theory of value" here we mean proving that the foundation of value — the centre of gravity around which prices oscillate — is labour time. If you are still puzzling over why celebrities are paid so much, why diamonds are so expensive, why downtown housing is so dear — questions of supply and demand and rent theory — you may first look at some answers and articles on Zhihu about the labour theory of value^{[4][5][6][7][8]}.

Proving the Labour Theory of Value

For classical political economists such as Smith, Ricardo, and Marx, the labour theory of value was self-evident, needing no argument. But after the marginalist revolution, marginal utility became the economic mainstream, and people stopped discussing the "substance of value" underlying prices. In the twentieth-century economic debates, both the defenders and the opponents of the labour theory of value argued from pure theory; no one had ever carried out a genuine empirical proof.

The reason is that commodity production in modern society is extremely complex. Tracing any single commodity back along the production chain of its inputs may involve the labour of thousands of different people. Without the input-output relations of these commodities, empirical verification is impossible. It was only after the Second World War, with the input-output tables of Leontief^{[9][10]}, that the means existed. These tables came to be valued by governments and were widely used by the US government and other Western developed-country governments. The input-output tables compiled by governments were used by economic management departments to guide the formulation of economic plans and industrial plans. From the 1980s, some Marxist economists recognised the theoretical value of the input-output tables and began to use them for empirical verification of the labour theory of value.

On quantitative empirical research on the labour theory of value, a paper by an economics professor at the University of Massachusetts^[11] is worth reading:

A third response appears in the work of Anwar Shaikh, who continues to use the standard interpretation of the labour theory of value. His main claim is similar to Ricardo's: values and prices are approximately equal at the level of the individual commodity. In his 1984 paper, Shaikh presented a theoretical argument showing that the deviation between prices and values is "small," and then used data from the Italian and US economies to show that his view was empirically correct. Shaikh's research on this question has generated a substantial literature, which I will review in the section on "The Classical Theory of Relative Prices" below. A fourth response comes from the work of two mathematicians, Emmanuel Farjoun and Moshé Machover, who introduced probabilistic methods into political economy. They argue that most economic variables, such as prices, profit rates, and wages, are non-degenerate random variables, each with its own probability distribution function. This implies that the equilibrium of the capitalist economy should be characterised by a distribution of profit rates, rather than a single, uniform profit rate. From a probabilistic point of view, the Sraffa-based critique rests on a false assumption, namely a uniform profit rate as the characteristic of long-run equilibrium; its conclusions are therefore invalid. Besides rebutting the Sraffa-based critique, the two authors also put forward an empirical theory of the distribution of profit rates and value-price deviations.

Anwar Shaikh represents the orthodox tradition of Marxist political economy: he insists on the assumption of a general "rate of profit," insists on the Marxian transformation of values into prices of production, and carries out empirical proof within this theoretical framework. Shaikh may be counted the founder and chief contributor of empirical research on the labour theory of value, and has influenced a great many Marxist economists.

On Shaikh's research, see chapter fifteen, "The Empirical Strength of the Labour Theory of Value," of his 1998 book *Marxian Economics: A Reappraisal*^[12]: In this book, Shaikh verified the labour theory of value and the theory of prices of production on the basis of publicly released historical input-output data, and concluded that market prices are consistent with both labour values and prices of production.

In our empirical analysis we compared market prices, labour values and standard prices of production calculated from US input-output tables for 1947, 1958, 1963 and 1972 using data initially developed by Ochoa (1984) and subsequently refined and extended by others (Appendix 15.2). Across input-output years we found that on average labour values deviate from market prices by only 9.2 per cent, and that prices of production (calculated at observed rates of profit) deviate from market prices by only 8.2 per cent (Table 15.1 and Figures 15.2-3).

See also Shaikh's article translated last year by the WeChat public account "Democratic Planned Economy Review (DEPR)," "Speaking with Data: Can the Debate on Labour-Time Accounting Be Put

to Rest?"^[13]: In this article, Shaikh, using the 2002 US input-output table for 403 sectors, proved that the labour theory of value of Volume I and the "prices of production" of Volume III of Capital, that is, of the Sraffian system, accord closely with the empirical data. At the same time, drawing on Sraffa's newly released archives, Shaikh rebutted the view that Sraffa and Marx stood opposed, arguing that in theory they are in fact consistent.

In the archives, Sraffa stressed that Marx was "of course" right — "statistical compensation in large numbers" would make the sum of prices on the empirical plane essentially identical to the sum of labour values. This is precisely what we find in our aggregates of circulating capital, variable capital, surplus output, value added, the rate of surplus value, the rate of profit, and the value-capital ratio (measured respectively at prices proportional to labour values, i.e. direct prices, and at Sraffa prices corresponding to the actual rates of profit, i.e. prices of production). In both the circulating-capital model and the fixed-capital model, Marx and Sraffa are both completely right. It follows that a large number of similar, representative industries should exhibit such regularities.

Farjoun and Machover, in their 1983 book *Laws of Chaos*, by contrast, comprehensively reject the assumption of a general rate of profit, thereby overturning Marx's "transformation problem" of values into prices of production in Volume III of Capital. Being mathematicians, familiar with the application of probabilistic methods in the natural sciences, they borrowed the methods of statistical mechanics: the economic agents of capitalist society, like the particles in air or water, can move in all sorts of ways (that is, they have a large number of degrees of freedom), yet taken as a whole they exhibit the regularities of macroscopic physical quantities. Their work not only opened a new direction for empirical research on the labour theory of value but in fact founded Marxist econophysics.

After Shaikh, many followed in his footsteps, verifying the theory with the data of various countries^{[14][15][16][17]}. Li Minqi and others verified the labour theory of value and the theory of prices of production with Chinese input-output tables^[18].

Laws of Chaos follows the simple labour theory of value of Volume I of Capital, introduces probabilistic methods, and verifies through empirical data. The book has not yet been translated into Chinese; the main points it proves with empirical data are summarised as follows.

Persistent dispersion: profit rates are not uniformly distributed but display persistently large dispersion, rather than tending toward a single average. Shape of the distribution: the distribution of profit rates is non-normal, single-peaked and tent-shaped (a Laplace distribution), not a normal distribution. Structural negative profits: negative profit rates are a structural, permanent feature of the economy, not a temporary anomaly. Rejection of the equilibrium hypothesis: "uniform profit rates" is an invalid and unrealistic assumption; instead, economic laws should be treated probabilistically, like statistical mechanics. Low profitability: a considerable share of firms (for example, 60.5% in their

study) earn profit rates below the average bank rate. Probabilistic law: the labour theory of value is not a strict direct price-value correspondence for each commodity but a statistical phenomenon, a "law of large numbers" phenomenon. Labour values as centre of gravity: although prices fluctuate constantly, the total labour time required to produce a commodity (direct labour time plus indirect labour time) determines its long-run average price. Verification of key concepts: their research supports the view that labour is the source of value and surplus value is identical with profit, even setting aside the complications of the "transformation problem."

The heretical econophysics school likewise agrees in trying to show that the simple labour theory of value based on probability theory has more explanatory power than the theory of prices of production. Cockshott's paper of 1997 concludes that the labour theory of value fits the actual data better than the theory of prices of production:

Most interpretations of value theory contain the expectation that Sraffian prices of production should predict actual prices more accurately than the simple labour theory of value (LTV). We find that this is not the case. On correlation measures, the Sraffian predictions perform about the same as the LTV, and only slightly better on entropy measures. This finding calls for an explanation. The fact that the Sraffian predictions fail to markedly outperform the LTV can be understood as follows: contrary to Sraffian theory, industries with a higher organic composition of capital tend to have lower rates of profit. Source: W. Paul Cockshott and Allin Cottrell, "The Scientific Status of the Labour Theory of Value"^[19]

Nils Fröhlich's research based on German data likewise proves, from the relationship between profit rates and the organic composition of capital, that the simple labour theory of value makes more sense than the theory of prices of production:

The results of this paper are similar to those of previous studies on labour values and market prices. The paper argues that both the labour theory of value and the neo-Ricardian theory of value achieve very good results in explaining the data. The differences between the estimates are practically negligible. Therefore, considering the methods proposed by Shaikh (1984) and Farjoun and Machover (1983), and bearing Occam's razor in mind, we should prefer the labour theory of value for the analysis of real phenomena. Moreover, there is a crucial point against the neo-Ricardian theory of value: the foundation of the transformation problem appears to be mistaken, since profit rates and capital intensity are negatively correlated. Furthermore, there are indications that profit rates follow a gamma distribution and are therefore not uniform. The corresponding density function changed little between 2000 and 2004. From the perspective of probabilistic political economy, the German economy thus appears to be in a state of statistical equilibrium. After all, although most classical scholars deny the relevance of labour values to the explanation of prices, there is good reason to hold that the law of value holds in a stochastic sense. Needless to say, this also means that the famous Marxian invariance assumptions can be proved in a similar fashion. Profits are therefore not based on the marginal

product of capital, but on exploited labour. Source: Nils Fröhlich, "Labour values, prices of production and the missing equalization of profit rates: Evidence from the German economy"^[20]

Because the empirical research on the labour theory of value involves a great deal of mathematical apparatus from linear algebra and probability theory, I will not go into it here; interested readers may find the papers themselves.

But as for the transformation problem, whichever school is right, or if all have some merit, it does not affect our conclusion: the labour theory of value is a scientific theory that has passed empirical testing. We can build on the labour theory of value, combined with various statistics, to predict the historical fate of capitalism.

Notes

[1] "The Labour Theory of Value and Marginal Utility Theory: Which Is a Scientific Economic Theory?" — It is interesting to compare the strong results of the simple labour theory of value with its main competitor, marginal value theory. Marginal value theory is based on the idea that prices evolve to levels at which marginal utilities per dollar are equalised across different products. This is an unfalsifiable proposition. Since subjective utility is unobservable, it is impossible to compare a country's price structure with utilities, and therefore impossible to carry out the kind of correlation study done for the labour theory of value. Any observed price structure can be said to reflect subjective utility. The problem is that, on marginal utility theory, subjective utility manifests itself only in the price structure people are willing to pay for commodities. In this case, one explains the relative prices of all currently traded commodities by a larger number of unknown variables — the subjective utility of each product to each member of the population (given current wealth and possession of other goods). The theory asserts that every person, in his current consumption pattern, derives the same marginal utility from a pound spent, across all products. But how do we know this?

中文原文：把简单劳动价值论这一有力的结果与其主要竞争者——边际价值论——作对比是很有意思的。边际价值论基于这样的思想，即价格会演进至每美元的边际效用在不同产品之间平均化后的水平（prices evolve to levels at which marginal utilities per \$ are equalised across different products）。这是一个无法证伪的命题。由于主观效用是无法观察发现的，无法将一个国家的价格结构与效用进行比较，所以不可能进行对劳动价值论所做的那种相关度研究。人们观察发现到任何价格结构，都可以说是反映了主观效用。问题在于，根据边际效用论，主观效用只有在人们愿意支付商品的价格结构中才能表现出来。在这种情况下，人们是用更大数量的未知变量——每个产品对人口中每个成员的主观效用（在给定当前的财富和拥有其他商品的情况下）——来解释所有当前交易商品的相对价格。这一理论断言，每一个人在他当前的消费模式下，花一英镑所得到的边际效用，对于所有产品都是相同的。但是我们怎么知道这一点呢？-----《劳动价值论与边际效用论：哪一个科学的经济理论？》

<https://review.youngchina.org/archives/14254>

[2] Hu Daiguang: "Sraffa's Theory of Commodity Production and Price Determination," Journal of Peking University (Philosophy and Social Sciences), 1985, No. 3 — Equations (9) and (11) provide us with $n+1$ equations and $n+2$ variables (n prices, the rate of profit r , and the wage W). The system therefore has one degree of freedom: once one

variable is fixed, the others are determined. If $W=1$, the entire national income goes to wages, meaning profit = 0, i.e. $r=0$. If $W=0$, r reaches its maximum. In the former case we return to equation (2), except that labour quantities are now explicitly expressed rather than the quantities of subsistence necessities. In the latter case we return to equation (7), except that, regardless of whether (7) includes consumer goods, the problem Sraffa particularly attends to is: if the wage level is known, the rate of profit and the prices of all commodities can be determined.

中文原文：方程(9)和(11)向我们提供 $n+1$ 个方程和 $n+2$ 个变量(n 个价格,利润率 r 和工资 W)。因此,这个体系的演算有一个自由度,只要有一个变量的数值被确定,其他变量就被确定了。如果 $W=1$,则全部国民收入就归于工资,这即意味着利润=0,即 $r=0$ 。如果 $W=0$,则 r 就达到最大。在前一种情况下,就回到(2)式,不同之处在于现在明确地表明劳动数量,而不用生存必需品数量来表示。在后一种情况下,就回到(7)式,不同之处在于不管(7)式包括消费品,斯拉法所特别注意要解决的问题是,如果工资水平为已知,那么就可确定利润率和所有商品的价格。--胡代光：《斯拉法的商品生产和价格决定理论》，北京大学学报(哲学社会科学版)一九八五年第三期

<https://ccj.pku.edu.cn/Article/Download?id=271015599&&type=ArticleFile>

[3] Roemer stresses that real wages in actual capitalist society can depart completely from Marx's subsistence wage; the struggle between the working class and the capitalist class, not the value of labour power, determines their movement. Therefore "reconstructing a Marxist theory of value without the concept of the subsistence wage is a possible path toward developing a theory of exploitation no longer based on the labour theory of value as a theory of exchange, and it will also become a necessary reconstruction." Combining Roemer's views with the foundational place of the labour theory of value in Marx's economic system as a whole, Roemer not only achieved the goal of "expressing his understanding of all aspects of Marxist economic theory through the construction of specific models," but also completed a thorough "remodelling" of Capital at the price of abandoning the labour theory of value.

中文原文：罗默强调资本主义现实社会中的真实工资完全可以脱离马克思的生存工资水平，工人阶级和资本家阶级之间的斗争而非劳动力的价值决定了它的变动，因此，“重建一个没有生存性工资概念的马克思主义价值理论，对于发展出一个不再建立在作为交换理论之劳动价值理论基础上的剥削理论来说，是一个可能的途径，而且它也将成为一个必要的重建”。综合罗默的上述观点，再结合劳动价值论在整个马克思经济理论体系中的基础性地位，罗默不仅实现了“通过建立一些特定的模型”来表达其对“马克思主义经济理论的各方面的理解”的目标，而且还以放弃劳动价值论为代价完成了对《资本论》的彻底“改造”。

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中文原文：我生产一瓶酒，然后把它埋了，十年后我没有付出额外的劳动时间，如何用劳动价值论来解释酒增值了？- 天马的回答 - 知乎 <https://www.zhihu.com/question/385767141/answer/1247274248>

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